

Education

University of Connecticut

PHD IN STATISTICS

Connecticut, USA

Sep 2018 - Sep 2023

University of Wisconsin - Madison

MASTER IN STATISTICS

Wisconsin, USA

Sep 2016 - May 2018

Nanjing University

BACHELOR IN STATISTICS

Jiangsu, China

Sep 2013 - Jun 2017

Work Experience

Quantitative Trading Book in Ernst & Young U.S. LLP

SENIOR CONSULTANT

New York, USA

Oct 2023 - Present

- **CCAR Stress Testing for Trading Book (ETD & OTC)**
 - Designed and implemented a CCAR stress testing framework to quantify P&L impacts for equity, FX, and commodity derivatives under regulatory stress scenarios.
 - Built trade-specific object-oriented classes to handle heterogeneous transaction structures across ETD and OTC products.
 - Integrated firm-wide front-office pricing systems with a custom fallback pricing engine to ensure robustness under data or system limitations.
 - Produced transparent and intuitive stress attribution reports, decomposing total P&L into individual stress factor impacts for senior risk stakeholders.
- **Value-at-Risk (VaR) Risk Matrix Construction and Maintenance**
 - Maintained and enhanced a VaR calculation framework by regularly updating risk matrices used across the trading book.
 - Cleaned and filtered historical time series data, organizing risk drivers by factor classes including equity, FX, interest rates, commodities, ABS, and MBS.
 - Estimated factor-level volatilities and cross-factor correlations to construct coherent covariance structures.
 - Computed trade-level VaR by combining factor sensitivities, volatilities, correlations, and MPOR, enabling consistent risk aggregation across portfolios.
- **Generative AI for Ongoing Performance Assessment (OPA) Reporting**
 - Developed a multi-agent Generative AI system using **LangGraph** to automate OPA validation report generation for trading book models.
 - Orchestrated specialized AI agents responsible for data ingestion, data cleaning, trade pricing, result validation, sensitivity testing, and executive-level summarization.
 - Enabled agents to dynamically select analytical tools from third-party **MCP libraries** and retrieve relevant inputs through a **RAG-based data access layer**.
 - Automatically generated end-to-end OPA reports, significantly improving consistency, transparency, and review efficiency for model risk management teams.
- **Optimization of American Options Pricing**
 - Applied the **American Monte Carlo (AMC)** method to replace a nested Monte Carlo approach for pricing American-style options.
 - Reduced computational complexity from $O(n^2)$ to $O(n)$, achieving substantial runtime improvements while preserving pricing accuracy.

Bank of China International Holdings Limited

SECURITIES ANALYST ASSISTANT (INTERN)

Shanghai, China

Jun 2021-Sep 2021

- Focused on battery and new energy industry. Predicted the short- and long-term performance of stocks of related companies based on time series model with a spike-and-slab error.
- Adjusted the prediction under a multinomial model based on the performance of correlated companies and avoided making an over-optimistic forecast compared with previous model.

HUATAI SECURITIES CO., LTD. (HTSC)

DATA ANALYST (INTERN)

Jiangsu, China

Jul 2017-Sep 2017

- Unsupervised screened visitors with a strong desire to buy products based on their records on company's APP.
- Cleaned and reshaped the 17 million visitor records by summarizing operations from the same visitor.
- Extracted useful variables by PCA (principal component analysis) method.
- Divided visitors into five groups by K-means methods and assigned visitors labels by their group.
- Fitted a decision tree with labeled data which could tag new visitor within 20 seconds while the target is 1 min.

Skill

- **Language:** Mandarin Chinese (Native), English
- **Coding/Database Languages:** Master R, Python, GitHub, Latex, Nimble, JUGS and HPC, familiar with SQL, SAS, MATLAB, C++ and Julia.
- **Certificate:** CFA level 1